

Brandon Yifan Yang

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EDUCATION

-  **University of Pennsylvania** Philadelphia, PA
M.S.E. in Robotics; GPA: 3.7/4.0 May 2027
-  **University of Virginia** Charlottesville, VA
B.S. in Computer Science, Highest Distinction; GPA: 3.9/4.0 May 2025
- **Selected Coursework:** Physical Intelligence, Learning in Robotics, GPU Programming & Architecture, Reinforcement Learning, Probabilistic ML, Natural Language Processing

EXPERIENCE

Research Lead San Francisco, CA
AfterQuery (YC W25) May 2026 - Present

- Led development of custom UMI-style hardware and a distributed robotics data-collection system for robot learning, spanning device-side capture, server-side processing, storage, and dataset management.
- Directed a 6-person research team building an AI-agent search benchmark, with task authoring and validation workflows, rubric-based grading, agent-run orchestration, trace logging, and reproducible model comparisons.

Embodied Intelligence Research Intern Beijing, China
Spirit AI May 2025 - Aug 2025

- Trained, evaluated, and deployed VLA model variants in PyTorch across simulation and a dual-arm robot, studying contrastive objectives, observation-noise curricula, reduced guidance inputs, and mixture-of-experts architectures.
- Designed and implemented a DAgger data-collection pipeline for VLA models, including operator-takeover logic and corrective-demonstration workflows used for model training and evaluation.
- Unified robotics simulation, training, and evaluation workflows into a reproducible Docker/Slurm pipeline for VLA experiments; open-sourced code and datasets on [GitHub](https://github.com).

RESEARCH EXPERIENCE

Perception, Action, & Learning Lab, University of Pennsylvania Philadelphia, PA
Advisor: Dinesh Jayaraman Aug 2025 - Present

- Implemented demonstration retrieval for $\pi_{0.5}$ VLA policies, integrating robot data retrieval into training and test-time inference to improve tabletop manipulation. (*CoRL 2026*)
- Engineered scalable data transformation, training, and evaluation pipelines using Docker, Slurm, Ray, JAX, PyTorch, and Hugging Face to support retrieval-based VLA policy variants.
- Designed and implemented infrastructure for collecting VLA hidden activations and evaluating activation-steering interventions across MetaWorld, LIBERO, RoboCasa, and real Franka robot experiments. (*ICML 2026 Workshop; NeurIPS 2026*)

Learning and Interactive Robotics, University of Virginia Charlottesville, VA
Advisor: Yen-Ling Kuo Aug 2024 - May 2025

- **Interpretable Vision-Language-Action Models via Skill Conditioning**
 - * Led *SkillVLA*, a skill-conditioned VLA model for language-conditioned manipulation with improved action interpretability via subgoal instructions and learned skill library.
 - * Presented as an oral talk at the 2024 UVA LLM Workshop ([slides](#)), earning the Audience Choice Award (top 3 of 28 presentations).
- **Contrastive Learning for Robot Manipulation**
 - * Performed contrastive learning over action sequences to learn behavior-grounded visual embeddings, improving visuomotor policies under heterogeneous camera poses and object appearances. (*CoRL 2025*)

- * Modified simulation environments and trained PyTorch-based visuomotor policies to evaluate learned embeddings.

Research Assistant, Collaborative Robotics Lab, University of Virginia

Advisor: Tariq Iqbal

Charlottesville, VA

May 2022 - May 2024

- **Grounded Location for Object Manipulation (GLOMA)** [code]
 - * Led team of 3 to develop zero-shot image-editing model grounded by language instructions for object relocation and manipulation tasks, designed for downstream robotic applications using goal-conditioned RL and Behavioral Cloning (BC).
 - * Integrated language grounding with visual perception by using bounding box guidance from pre-trained language models, enabling precise object relocation without external supervision and improving baseline performance by 65%.
 - * Collected and annotated custom dataset for fine-tuning pre-trained language and vision models.
 - * Presented poster at 3 conferences and symposiums.
- **Centralized multi-agent RL for Collaborative Tasks** [code]
 - * Developed long-horizon on/offline centralized MARL for robotic bolt screwing tasks.
 - * Designed and optimized custom reward functions in multi-agent framework for task completion and agent collaboration, improving task success rate by 20%.
 - * Deployed and tested custom simulated environments in IsaacGym for training and evaluation.

PUBLICATIONS

Jiani Huang*, **Brandon Y. Yang***, Amish Sethi, Yuchen Zheng, Christopher Watson, Jianing Qian, Mayur Naik, and Dinesh Jayaraman. “Retrieval-Augmented Vision-Language-Action Model.” Submitted to *CoRL 2026*.

Miranda Muqing Miao*, Subin Kim*, **Brandon Y. Yang***, Dinesh Jayaraman, and Lyle Ungar. “Contrastive Conceptor Activation Steering (COAST): Unlocking Vision-Language-Action Models through Hidden States.” *ICML 2026 Mechanistic Interpretability Workshop; NeurIPS 2026*. [Workshop][arXiv]

Amish Sethi*, **Brandon Y. Yang***, Michael Evans, Spencer Mateega, and Sam Jung. “Position: Trust No Multi-Turn Benchmark Without Measuring Its User Simulator.” Submitted to *NeurIPS 2026 Position Paper Track*.

Lee, Sung-Wook, Xuhui Kang, **Brandon Y. Yang**, and Yen-Ling Kuo. “Class: Contrastive learning via action sequence supervision for robot manipulation.” *Conference on Robot Learning*, pp. 4743-4766. PMLR, 2025. [Website][arXiv][PDF]

*Equal contribution.

SOFTWARE PROJECTS

openpi-cuda [GitHub]: Developed custom CUDA kernels via C++/Python bindings to accelerate $\pi_{0.5}$ VLA model inference; achieved 18.5ms latency reduction over baseline PyTorch.

notie-markdown [Website]: Markdown rendering web app with support for equation previewing, graphing, code running, and more. Built with React, TypeScript, Python, Flask. Deployed on Vercel and Heroku.

Blogs and Notes [blog, notes]: Detailed blog posts and course notes with visualizations, graphs, and code on CS and AI topics, written with notie-markdown.

SmartOH [GitHub]: Queue management system designed for office hours. Features real-time queue updates, notifications, and analytics. Built with Next.js, TypeScript, Python, FastAPI. Deployed with CI/CD pipeline on Vercel and AWS EC2.

Voy: Collaborated with 7 non-profits to develop Voy, a volunteer and driver management platform; competed and received \$1000 in funding from UVA’s Entrepreneurship Cup. Built with React, Node.js, TypeScript, JavaScript.

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript

Tools & Frameworks: uv, ruff, ty, Claude Code, Codex, PyTorch, Jax, CUDA, Slurm, Ray